QRange project passport

Question

The author needed to implement a memory management system during the execution of mathematical functions. The occupied memory represented mathematical ranges in hexadecimal notation. This task involved relatively straightforward logic, prompting the author to search for libraries or ready-made solutions for working with mathematical ranges in C++ using Qt.

Analogues	Features
std::ranges	filtering, transformation
Boost.Range	adapters and algorithms
Range V3	functional programming
think-cell public library	graphical representation

Problem

Undoubtedly, the libraries "std::ranges," "Boost.Range," "Range V3," and "the think-cell public library" represent powerful solutions for working with ranges. However, it is important to consider the following points:

1. Each of these libraries is primarily designed for discrete collections (such as vectors or lists) rather than continuous ranges.
2. Some of these libraries may introduce overhead due to their levels of abstraction.
3. The syntax and semantics of these libraries can be complex, particularly for users who are not familiar with modern C++ features.

In summary, there are three main drawbacks: the concept of ranges is approached from the perspective of collections of values, the complexity of the libraries, and the potential overhead associated with their use.

Actuality

At present, the author has not identified a concise and easily integrable solution for tasks requiring work with mathematical ranges within a project developed in C++ using the Qt framework.

Purpose

The objective of the QRange project is to develop a robust and user-friendly library for handling mathematical ranges. This library will enable developers to efficiently and effortlessly perform operations involving ranges, such as union, intersection, membership verification, partitioning, and more. The library is designed to be intuitive and well-documented.

The utilization of QRange is expected to reduce the time developers spend learning the library by 45-50% compared to existing alternatives, while also enhancing the user experience (UX) by 15-20%. The effectiveness results will be described in greater detail after a period of six months.

Tasks

- Implement the project
- Add unit tests
- Write documentation
- Investigate and analyze the effectiveness and usability of the QRange

About

Name:	Author:	Version:	License:	Date:
QRange	25-masik-52	1.0.0	MIT	11.01.2025

Realization:	Documentation:	Unit-tests:
Using C++17 and the Qt framework.	Doxygen	Google Test